DIFFUSION STOCHASTIC LEARNING OVER MULTI-TEAM NETWORK GAMES

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Abstract

This work addresses stochastic multi-network competing problems. Unlike prior studies that focus exclusively on the two-team (or two-network) case, we consider the more general setting with $T \geq 2$ teams. In the multi-team setting, each team consists of a set of partially connected cooperative agents, and there exists at least one link between any two teams to enable information exchange. We propose a new algorithm based on a diffusion learning strategy, which subsumes the existing two-team competing algorithm as a special case. We show that the state vectors generated by the learning algorithm asymptotically converge to a neighborhood of the Nash equilibrium in the sense of first-, second-, and fourth-order error moments.

Introduction

- Teams of agents are formed. Agents cooperate within their teams but compete against those in the opposing teams to optimize the cost function of the team.
- Each agent learns its strategy by interacting with its immediate neighborhood.
- The work develops a multi-team competing algorithm and studies its stability and convergence.

Game Formulation

1. Team structure: A total of K agents with index set $\mathcal{N}$ is partitioned into mutually disjointed teams $\mathcal{N}^{(1)}, \ldots, \mathcal{N}^{(T)}$.

2. Game formulation: Each agent $k \in \mathcal{N}^{(t)}$ aims to optimize the following cost function

$$J_k^{(t)}(x) := \mathbb{E}_{\boldsymbol{\xi}_k}\left[Q_k^{(t)}(x^{(1)}, \ldots, x^{(T)}; \boldsymbol{\xi}_k)\right].$$

- $x^{(t)}$ is the strategy of team t.
- $x = \mathrm{col}\{x^{(t)}\}_t$ is the collection of all strategies.
- $\{\boldsymbol{\xi}_k\}$ are local random samples.
- Agents within the same team pursue a common optimization objective.

The m-ATC-ITC Algorithm

An agent $k \in \mathcal{N}^{(t)}$ updates its strategy $x_k^{(t)}$ by leveraging its estimates on the strategies of opposing teams $x_k^{(\tau)}$ $(\tau \neq t)$ via the following two steps:

1. ATC (strategy update): Let $A^{(t)} = [a_{\ell k}^{(t)}]$ be left stochastic and primitive,

$$\boldsymbol{x}_{k,i}^{(t)} = \sum_{\ell \in \mathcal{N}^{(t)}} a_{\ell k}^{(t)} \left(\boldsymbol{x}_{\ell,i-1}^{(t)} - \mu^{(t)} \widehat{\nabla_t J_\ell^{(t)}}(\boldsymbol{x}_{\ell,i-1})\right).$$

- First adapt then combine in-team information.
- $\mu^{(t)}$ is the learning rate of team t.
- ∇_t denotes the partial gradient applied with respect to the argument $x^{(t)}$.
- The $\widehat{(\cdot)}$ denotes the stochastic gradient.

2. ITC (estimate update): Let $C^{(\tau t)} = [c_{\ell k}^{(\tau t)}]$ and $[C^{(tt)}; C^{(\tau t)}]$ be left stochastic,

$$\boldsymbol{x}_{k,i}^{(\tau)} = \sum_{\ell \in \mathcal{N}^{(t)}} c_{\ell k}^{(tt)} \left(\boldsymbol{x}_{\ell,i-1}^{(\tau)} - \mu^{(\tau)} \hat{\boldsymbol{g}}_\ell^{(\tau)}(\boldsymbol{x}_{\ell,i-1})\right) + \sum_{\ell \in \mathcal{N}^{(\tau)}} c_{\ell k}^{(\tau t)} \boldsymbol{x}_{\ell,i}^{(\tau)}.$$

- First infer the state of opposing teams, then combine in-team and out-of-team information.
- $g_\ell^{(\tau)}$ denotes the local knowledge agent $\ell \in \mathcal{N}^{(t)}$ has about team τ, that is, $g_k^{(\tau)}(x) := \nabla_\tau J_{\ell_k}^{(\tau)}(x)$ for some correspondence $\ell_k \in \mathcal{N}^{(\tau)}$.

Theoretical Analysis

Definition 1 (Nash equilibrium of the game) *Let $p^{(t)}$ be the Perron vector of $A^{(t)}$ with $p_k^{(t)}$ being its kth element. Define*

$$J^{(t)}(x) = \sum_{k \in \mathcal{N}^{(t)}} p_k^{(t)} J_k^{(t)}(x).$$

Then the decision vector $x^\star = \mathrm{col}\{x^{(t),\star}\}_{t \in [T]}$ is the Nash equilibrium of the network game if the following condition holds: $\forall\, t \in [T], \forall\, x^{(t)}$,

$$J^{(t)}(x^\star) \leq J^{(t)}\left(x^{(t)}, x^{-(t),\star}\right).$$

To analyze the evolution of states across agents, we denote the following concatenated vector

$$\boldsymbol{X}_i := \mathrm{col}\left\{\mathrm{col}\left\{\boldsymbol{x}_{k,i}^{(\tau)}\right\}_{k \in \mathcal{N}}\right\}_{\tau \in [T]}.$$

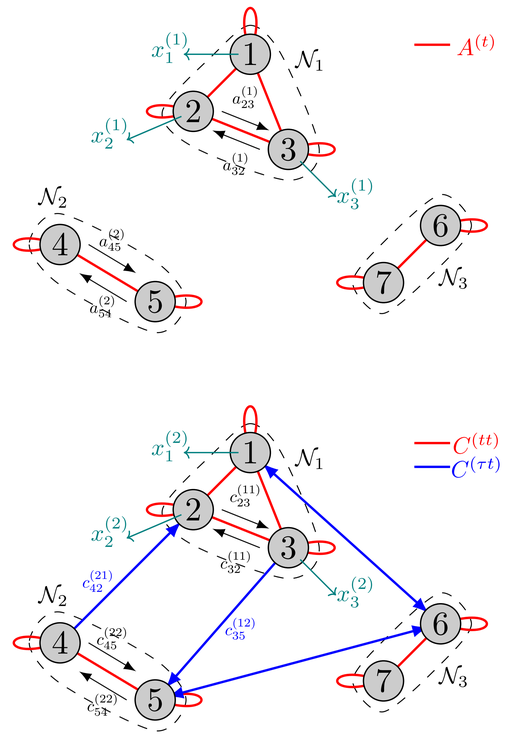

Figure 1: m-ATC-ITC algorithm. Top: ATC. Bottom: ITC.

The goal is to show that $\boldsymbol{X}_i$ converges to the Nash equilibrium, namely, $X^\star := \mathrm{col}\left\{\mathbf{1}_K \otimes x^{(\tau),\star}\right\}_{\tau \in [T]}$, where $\mathbf{1}_K \in \mathbb{R}^{K \times 1}$ is the vector of ones.

Theorem 1 (Network error moment stability) *Consider a network of K agents forming T teams running the m-ATC-ITC algorithm. Under some mild assumptions and for sufficiently small step sizes $\mu^{(t)}$, the network converges asymptotically to a neighborhood around the Nash equilibrium $X^\star$. That is, the error state $\widetilde{\boldsymbol{X}}_i := X^\star - \boldsymbol{X}_i$ is stable in the sense of*

$$\limsup_{i \to \infty} \mathbb{E}\left[\left\|\widetilde{\boldsymbol{X}}_i\right\|^2\right] = O(\mu),$$

$$\limsup_{i \to \infty} \mathbb{E}\left[\left\|\widetilde{\boldsymbol{X}}_i\right\|^4\right] = O(\mu^2),$$

$$\limsup_{i \to \infty} \left\|\mathbb{E}\left[\widetilde{\boldsymbol{X}}_i\right]\right\| = O(\mu).$$

Simulation Results

We consider a quadratic game with local losses of the form

$$J_k^{(t)}(x) = \mathbb{E}\left[(x^{(t)})^\mathsf{T} \boldsymbol{Q}_k x + \boldsymbol{L}_k^\mathsf{T} x^{(t)}\right].$$

with $\mathcal{N}^{(1)} = \{1,2,3\}$, $\mathcal{N}^{(2)} = \{4,5\}$, and $\mathcal{N}^{(3)} = \{6,7,8\}$, we have the following results by averaging over 200 samples.

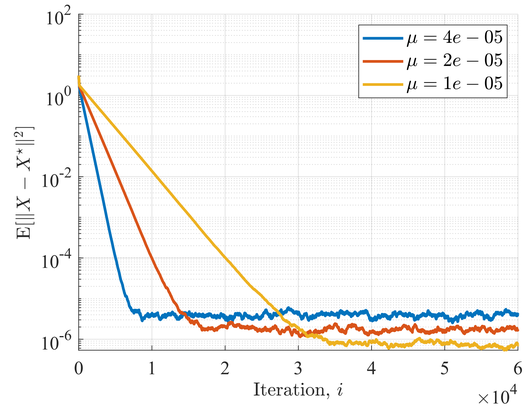

Figure 2: Second moment stability. The limit superior is numerically proportional to $\mu^{1.18}$.

Concluding Remarks

- We proposed a diffusion learning algorithm for multi-network competing problem.
- We analyzed the stability of the proposed algorithm by showing its convergence towards the neighborhood of the Nash equilibrium.
- The proof of the theorem is inspired by the work on traditional distributed optimization problem (see *Adaptation, Learning, and Optimization over Networks* by A. H. Sayed).
- The detailed proofs and the reproducible code can be found via the link `https://wenperng.github.io/publication/2026-ICASSP`.

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